

vector space

A vector space is a non-empty set V equipped with two operations "vector addition" $(+)$ and "scalar multiplication" $(\cdot)$ as follows.

vector addition

For $u, v \in V$,

$$\rightarrow u + v \in V \text{ (closure property)}$$

$$\rightarrow u + v = v + u \text{ (commutativity)}$$

$$\rightarrow (u + v) + w = u + (v + w) \text{ (associativity)}$$

$$\rightarrow u + 0 = u \text{ (existence of unique zero element in } V)$$

$$\rightarrow u + (-u) = 0 \text{ (existence of additive inverse in } V)$$

Scalar multiplication

For any scalar α, β ,

- $\alpha u \in V$ (closed under multiplication)
- $\alpha(u+v) = \alpha u + \alpha v$ (distributivity)
- $(\alpha+\beta)u = \alpha u + \beta u$ (")
- $\alpha(\beta u) = (\alpha\beta)u$ (associativity)
- $1 \cdot u = u$ (existence of multiplicative identity in V)

Note:

If the elements of V (i.e. $u, v, w, \dots$) and scalars $\alpha, \beta, \dots$ are real, then V is a real vector space otherwise a complex vector space.

Problem

① set of all polynomials of degree ≤ 2 , given by $V = \{ \underline{p(x) = a_0x^2 + a_1x + a_2} \mid a_0, a_1, a_2, a_3 \text{ are } \}$ real nos. $\}$ is a vector space.

Solution

Let $\underline{p(x), q(x), r(x) \in V}$, where

$$p(x) = a_0x^2 + a_1x + a_2$$

$$q(x) = b_0x^2 + b_1x + b_2$$

$$\rightarrow r(x) = c_0x^2 + c_1x + c_2.$$

$$a_0, a_1, a_2, b_0, b_1, b_2,$$

$$c_0, c_1, c_2 \rightarrow \text{are all real nos.}$$

$$\begin{aligned} \text{(i)} \quad p(x) + q(x) &= (a_0x^2 + a_1x + a_2) + (b_0x^2 + b_1x + b_2) \\ &= \underline{(a_0 + b_0)x^2 + (a_1 + b_1)x + (a_2 + b_2)} \\ &\quad \underline{\in V} \quad \text{of the form } a_0x^2 + a_1x + a_2 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad p(x) + q(x) &= (a_0x^2 + a_1x + a_2) + (b_0x^2 + b_1x + b_2) \\
 &= (a_0 + b_0)x^2 + (a_1 + b_1)x + (a_2 + b_2) \\
 &= (b_0 + a_0)x^2 + (b_1 + a_1)x + (b_2 + a_2) \\
 &= (b_0x^2 + b_1x + b_2) + (a_0x^2 + a_1x + a_2) \\
 &= q(x) + p(x)
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \{p(x) + q(x)\} + r(x) &= \{(a_0x^2 + a_1x + a_2) + (b_0x^2 + b_1x + b_2)\} \\
 &\quad + (c_0x^2 + c_1x + c_2) \\
 &= \underbrace{(a_0 + b_0)x^2 + (a_1 + b_1)x + (a_2 + b_2)}_{+ (c_0x^2 + c_1x + c_2)} \\
 &= [a_0 + (b_0 + c_0)]x^2 + [a_1 + (b_1 + c_1)]x \\
 &\quad + [a_2 + (b_2 + c_2)] \\
 &= (a_0x^2 + a_1x + a_2) + \\
 &\quad \{ (b_0 + c_0)x^2 + (b_1 + c_1)x + (b_2 + c_2) \} \\
 &= p(x) + \{q(x) + r(x)\}
 \end{aligned}$$

(iv) zero polynomial

$$\text{def } 0 = 0(x) = 0x^2 + 0x + 0 \in V$$

Identity

$$\begin{aligned} \text{(i2)} \quad p(x) + 0(x) &= (a_0x^2 + a_1x + a_2) + (0x^2 + 0x + 0) \\ &= a_0x^2 + a_1x + a_2 \\ &= p(x). \end{aligned}$$

(v) $\forall p(x) \in V, \exists -p(x) \in V \Rightarrow$

Inverse

$$\begin{aligned} p(x) + (-p(x)) &= (a_0x^2 + a_1x + a_2) + (-a_0x^2 - a_1x - a_2) \\ &= (a_0 - a_0)x^2 + (a_1 - a_1)x + (a_2 - a_2) \\ &= 0x^2 + 0x + 0 \\ &= 0 \end{aligned}$$

Let α, β be scalars.

$$\begin{aligned} \text{(i)} \quad \alpha \cdot p(x) &= \alpha (a_0 x^2 + a_1 x + a_2) \\ &= \alpha a_0 x^2 + \alpha a_1 x + \alpha a_2 \\ &\in V \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \alpha \{p(x) + q(x)\} &= \alpha \{ (a_0 x^2 + a_1 x + a_2) + (b_0 x^2 + b_1 x + b_2) \} \\ &= \alpha \{ (a_0 + b_0) x^2 + (a_1 + b_1) x + (a_2 + b_2) \} \\ &= \alpha (a_0 + b_0) x^2 + \alpha (a_1 + b_1) x + \alpha (a_2 + b_2) \\ &= (\alpha a_0 x^2 + \alpha a_1 x + \alpha a_2) + (\alpha b_0 x^2 + \alpha b_1 x + \alpha b_2) \\ &= \alpha p(x) + \alpha q(x). \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad (\alpha + \beta) p(x) &= (\alpha + \beta) (a_0 x^2 + a_1 x + a_2) \\
 &= (\alpha + \beta) a_0 x^2 + (\alpha + \beta) a_1 x + (\alpha + \beta) a_2 \\
 &= \alpha a_0 x^2 + \alpha a_1 x + \alpha a_2 + \beta a_0 x^2 + \beta a_1 x + \beta a_2 \\
 &= \alpha p(x) + \beta p(x).
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad (\alpha \beta) \cdot p(x) &= \alpha \beta (a_0 x^2 + a_1 x + a_2) \\
 &= \alpha \beta a_0 x^2 + \alpha \beta a_1 x + \alpha \beta a_2 \\
 &= \alpha \{ \beta a_0 x^2 + \beta a_1 x + \beta a_2 \}
 \end{aligned}$$

Multiplicative identity

For every $p(x) \in V$, $\exists 1 \in V \ni p(x) \cdot 1 = p(x)$

$$(v) \quad 1 \cdot p(x) = p(x)$$

existence of multiplicative inverse.

$$\begin{aligned}
 &0x^2 + 0x + 1 \leftarrow \\
 &\in V
 \end{aligned}$$

Hence V is a vector space.

② Let V be the set of all ordered pairs of real nos. Addition and multiplication are defined by

$$(x, y) + (x_1, y_1) = (x + x_1, y + y_1) \text{ and}$$

$$\alpha(x, y) = (x, \alpha y), \text{ where}$$

x, y, x_1, y_1 are real nos., then V is not a vector space.

Soln consider $(\alpha + \beta)(x, y) = (x, (\alpha + \beta)y)$
 $= (x, \alpha y + \beta y)$ ①

also consider

$$\alpha(x, y) + \beta(x, y) = (x, \alpha y) + (x, \beta y)$$
$$= (2x, (\alpha + \beta)y) \text{ — ②}$$

$$\Rightarrow (\alpha + \beta)(x, y) \neq \alpha(x, y) + \beta(x, y)$$

$\Rightarrow V$ is not a vector space.

Subspace

A nonempty subset S of a vector space V is called a subspace of V , if S itself is a vector space with rule of addition and scalar multiplication, same as on vectors of V .

Note: If S is a subset of V , but rule of addition or scalar multiplication or both are not same as for V , then S is not a subspace of V .

Theorem (X)

A nonempty subset S of a vector space V is a subspace of V , iff

(i) If $\forall u, v \in S$, $u+v \in S$.

(ii) If $\forall u \in S$ and $\forall$ scalar α , $\alpha u \in S$.

Problem ① Let $V = \mathbb{R}_3$ be the vector space as usual.

Let $U = \{u = (x_1, x_2, x_3) \in V_3 \mid x_1 + x_2 + x_3 = 0\}$. s.t

U is a subspace of V .

Solution Let $u, v \in U$

$u = (x_1, x_2, x_3)$ and $v = (y_1, y_2, y_3)$

$\Rightarrow x_1 + x_2 + x_3 = 0$ — ①

and $y_1 + y_2 + y_3 = 0$ — ②

consider

$$u + v = (x_1, x_2, x_3) + (y_1, y_2, y_3)$$

$$= (x_1 + y_1, x_2 + y_2, x_3 + y_3) \xrightarrow{\text{T.P}} \in U$$

$$= x_1 + y_1 + x_2 + y_2 + x_3 + y_3$$

$$= [x_1 + x_2 + x_3] + [y_1 + y_2 + y_3]$$

$$= 0 + 0 \quad (\text{using ① \& ②})$$

$$= 0 \quad \Rightarrow u + v \text{ also } \in U.$$

consider α - any scalar & $u \in U$, T.P αu also $\in U$.

$$\begin{aligned}\alpha u &= \alpha(x_1, x_2, x_3) \\ &= (\alpha x_1, \alpha x_2, \alpha x_3) \\ &= \alpha x_1 + \alpha x_2 + \alpha x_3 \\ &= \alpha(x_1 + x_2 + x_3) \\ &= \alpha(0) = 0\end{aligned}$$

$\therefore U$ - is a Subspace of V .

② let $V_3 = \{(x_1, x_2, x_3)\}$ be a vector space with usual addition and scalar multiplication. U - subset of V_3 , defined by $U = \{(x_1, x_2, x_3) \in V_3 / x_1 x_2 + x_2 x_3 = 0\}$

Solution

let $u, v \in U$. $u = (x_1, x_2, x_3) \Rightarrow x_1 x_2 + x_2 x_3 = 0$ (1)

$v = (y_1, y_2, y_3) \Rightarrow y_1 y_2 + y_2 y_3 = 0$ (2)

consider

$$u+v = (x_1, x_2, x_3) + (y_1, y_2, y_3)$$

$$= (x_1 + y_1, x_2 + y_2, x_3 + y_3)$$

$$\text{To show } \{x_1 + y_1\} \cdot \{x_2 + y_2\} + \{x_2 + y_2\} \cdot \{x_3 + y_3\} = 0.$$

$$\text{LHS } (x_2 + y_2) \{x_1 + y_1 + x_3 + y_3\}$$

$$(x_2 + y_2) \{x_1 + x_3 + y_1 + y_3\}$$

$$(i) (x_2 + y_2)(x_1 + x_3) + (x_2 + y_2)(y_1 + y_3)$$

$$(ii) x_2(x_1 + x_3) + y_2(x_1 + x_3) + x_2(y_1 + y_3) + y_2(y_1 + y_3)$$

$$(iii) 0 + y_2(x_1 + x_3) + x_2(y_1 + y_3) + 0 \quad (\text{using } \textcircled{1} \text{ \& } \textcircled{2})$$

$$y_2(x_1 + x_3) + x_2(y_1 + y_3) \neq 0 \text{ in genl.}$$

$$\Rightarrow u + v \notin U. \quad \therefore U - \bar{m} \text{ not a subspace of } V_3.$$

In particular,